MARKING SCHEME 2024 -25
Class X Science (086)

Section-A		
1	D. Red-coloured copper is oxidized to black coloured copper(II) oxide	1
2	C. Add another chlorine atom to accept an electron from the calcium atom.	1
3	D. Potassium carbonate	1
4	D. Both (A) and (B)	1
5	B. combination reaction as well as oxidation reaction.	1
6	C. 18	1
7	C. $BaCl_2$ solution	1
8	D. Glucose →(cytoplasm) Pyruvate→ Ethanol (cytoplasm) + Carbon dioxide	1
9	C. Valves ensure that the blood does not flow backwards	1
10	D. test tube B as HCl will activate pepsin for breakdown of protein into simple molecules	1
11	D. axonal end of one neuron to dendritic end of another neuron.	1
12	A. hydra and planaria	1
13	D. behind the mirror and it's position varies according to the object distance.	1
14	D. Device X is a convex lens and device Y is a concave mirror, whose focal lengths are 20cm and 25cm respectively.	1
15	B. By splitting of a molecular oxygen into free oxygen atoms in presence of high energy UV radiations.	1
16	C. biomagnification	1
17	C. A is true but R is false.	1
18	C. A is true but R is false.	1
19	D. A is false but R is true	1
20	C. A is true but R is false.	1

Section-B

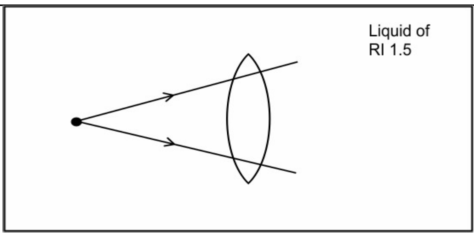
21 (2 marks)

S. No.	Feature	Alveoli	Nephron
1	Structure and location	Balloon like structures present at the terminal ends of bronchioles in lungs	Tubular structure present in kidneys
2	Function	Exchange of gases	Filtration of blood to form urine

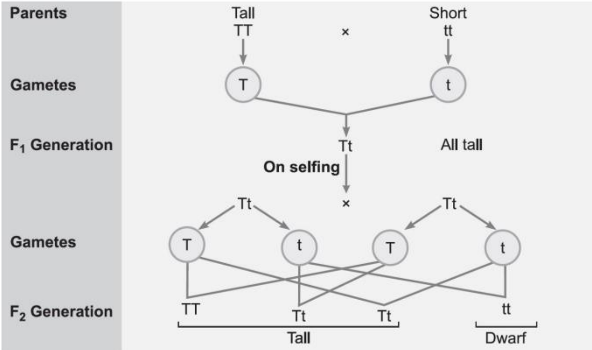
22 (2 marks)

A. $2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$

B. $Mg(OH)_2 + 2HCl \longrightarrow MgCl_2 + 2H_2O$

23	*Students to attempt either option A or B.* A. Release of carbon dioxide and intake of oxygen gives evidence that either photosynthesis is not taking place or its rate is too low. Normally during daytime the rate of photosynthesis is much more than the rate of respiration so carbon dioxide produced during respiration is used up for photosynthesis hence carbon dioxide is not released. **OR** B. The enzyme salivary amylase is present in our saliva present in the buccal cavity. Salivary gland produces the salivary amylase. In the absence of this enzyme, digestion of complex starch to simple sugar would be difficult, this affects the digestive process. Additionally, the swallowing of food would also be difficult as saliva makes the food wet so that it easily flows down the oesophagus.	2
24	 (Students should draw the rays through the lens and further in the liquid in the same direction as the incident ray without showing any bending at the interface of the liquid and the lens.) Justification: The light ray does not bend as the refractive index of both the mediums are same.	2
25	A. Phytoplankton, Producers will still have the highest amount of energy captured from sunlight which will continue to reduce as we move towards the top of the pyramid. B. Since the total mass in the lower trophic level is lesser there will be lesser food available to higher trophic levels causing organisms to die sooner than usual.	2
26	*Student to attempt either A or B.* A. The resistance will be lowest/minimum if all the resistors are connected in parallel. The equivalent resistance in parallel combination is given by (0.5) $\frac{1}{R_{eqv}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$ (0.5) Substituting the values, we get $\frac{1}{R_{eqv}} = \frac{1}{3} + \frac{1}{6} + \frac{1}{9} + \frac{1}{12}$ (0.5) $= \frac{12+6+4+2}{36} = \frac{24}{36} = \frac{2}{3}$ $R_{eqv} = \frac{3}{2} = 1.5\ \Omega$ (0.5) **OR** B. $P = V \times I$ (0.5) $I = \frac{P}{V} = \frac{1000}{220} = 4.54\ A$ (0.5) We will be using fuse B with is rated as 5A. This is because it will be able to sustain the current (4.54 A) passing through it. Whereas fuse A will melt and break the circuit as the current exceeds its rating. (1)	2

Section-C		
27	A. Aluminium (1/2) B.1 mark each for correct equations: $Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + H_2O$ $Al_2O_3 + 6HCl \longrightarrow 2AlCl_3 + 3H_2O$ C. It would displace iron to form aluminium oxide. (1/2)	3
28	*Students to attempt either A or B.* A. (i)The chemical name is Sodium carbonate. Its common name is Washing Soda. The chemical formula of the compound in its hydrated form is $Na_2CO_3.10H_2O$. (1) (ii) Sodium carbonate is prepared in the industries by various processes. One such common process is the thermal decomposition of sodium bicarbonate.The reaction that is involved here is as follows: (1) $NaCl+NH_3+CO_2+H_2O \rightarrow NaHCO_3+NH_4Cl$ $2NaHCO_3 \rightarrow Na_2CO_3+H_2O+CO_2$ Thus, in the manufacture process, we can find the use of brine (NaCl) in the production of sodium carbonate. (iii) As sodium carbonate is water-soluble and magnesium carbonate and calcium carbonate are insoluble, so it is used to soften water by removing $Mg^{2+}$and Ca^{2+}. Thus, when sodium carbonate is treated with water containing Ca and Mg salts, it removes them and thus, softens the water. (1) **OR** B. (i) As bee sting is acidic and wasp sting is basic. (1) (ii) To change the nature of soil to (neutral or basic). (1) (iii) To protect sculptures from the effects of certain gases present in environment and acid rain. (1)	3
29	To get rid of excretory products, plants use the following ways • Plants get rid of gaseous waste product through stomata present on leaves and lenticels on the stem. • The plants get rid of solid and liquid wastage by shedding of leaves, peeling of bark and felling of the fruits. • The plants also secret waste in the form of gums and resins. • Some waste substances are excreted through root into soil around them. • Excess of water is eliminated by the process of transpiration in living cell. (ANY 3 POINTS)	3

30	Mendel used garden pea for his experiment. (1)  (1)	
	When Mendel crossed a pure tall plant (TT) with a dwarf plant (tt), the progeny thus obtained was called F1 generation. Then he self-pollinated generation and found out 25% pure tall plants and 25% pure dwarf and 50% hybrid tall. F2 generation phenotypic ratio: 3:1 F2 generation genotypic ratio: 1:2:1 (1)	3
31	A. The near point and far point of the eye are determined with regards to the function of the lens of the eye. (1) B. In case of farsightedness, we use a convex lens or converging lens which helps light focus on the retina which is falling behind the retina in such away it corrects hypermetropia or far sightedness. (1) C. If a person is advised to wear spectacles with concave lenses, he is likely suffering from myopia, also known as short-sightedness or near-sightedness. (1)	3
32	A. Voltmeter measure the potential difference across two points and ammeter measure the current in the circuit (1.5) B. The resistance of a voltmeter should be high so that it takes a negligible current from the circuit. (1.5)	3
33	A. The magnetic field at P and Q is the same. Because the magnetic field lines inside the helical coil of wire which behaves like a solenoid is uniform/in the form of parallel straight lines. (1.5) B. Increasing/decreasing the number of turn in the coil. Increasing/decreasing the current through the coil. (1.5)	3

	Section-D	
34	*Students to attempt either option A or B.* A. (i) Vegetative propagation/ asexual reproduction. (0.5+0.5) (ii) More crops in same time interval, genetically identical, flower fruit faster, no need to depend on pollinators. (1+1) (iii) Cross pollination, the pollen from anther will be transferred the stigma of another banana plant using agents like wind, water, or insects etc. (0.5+0.5) (iv) There would be minor changes/some variation during the process of copying of the DNA. (1) **OR** B. Nutrients/glucose/oxygen/waste. (any two) (1) (i) less surface area for nutrients (glucose/oxygen) to pass from mother to embryo slow growth. (1) (ii) uterus; has thick lining with rich supply of blood to nourish the embryo.	5

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<td>(iii) (1)
a) male child
misused as if the foetus is female, some people engage in aborting the child leading to female foeticide. (2)</td>
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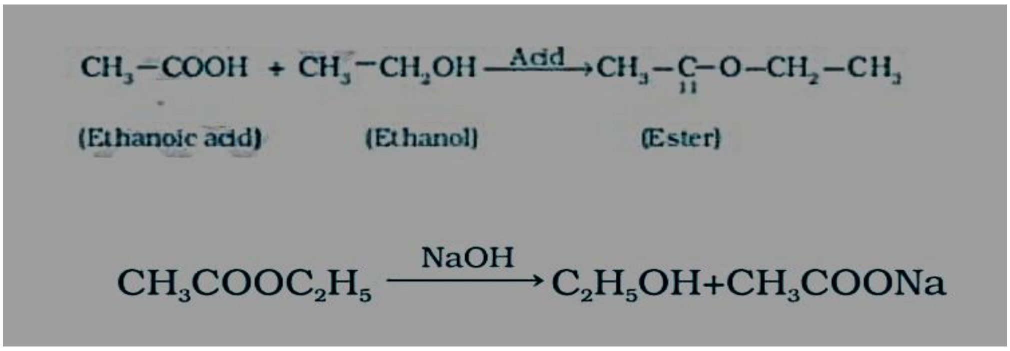
<td>35</td>
<td><u>Student to attempt either option A or B.</u>
A.(i)Rehmat’s observation is correct as the hydrogen atoms are substituted by hetero atom i.e., Cl.
$CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ (in presence of sunlight) (1)
OR
Any other relevant equation in the chain reaction
(ii) Electrolysis of brine
$2NaCl(aq) + 2H_2O(l) \longrightarrow 2NaOH(aq) + Cl_2 + H_2$
$NaCl \rightarrow Na^+ + Cl^-$
$2Cl^- \rightarrow Cl_2 + 2e-$ (At anode)
$H_2O \rightarrow H^+ + OH^-$
$2H^+ + 2e \rightarrow H_2$ (At cathode)
$Na^+ + OH^- \rightarrow NaOH$ [2]
Products: Sodium hydroxide/ NaOH/ Caustic soda [1]
And Hydrogen -
Uses: (any one each) [1]
Sodium hydroxide/ NaOH/ Caustic soda
- Degreasing of metals
- Preparation of soaps and detergents
- Paper making
- Artificial fibres
Hydrogen -
- Fuels
- Margarine
- Manufacture of ammonia for fertilizers
OR
B. X - Ethanoic acid/ acetic acid/ CH3COOH 3x1
Y - Ethanol/ Ethyl alcohol/ C2H5OH
Z - Ethyl ethanoate/ Ester – CH3COOC2H5


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36 Given: Resistance, $R1=R2=15\Omega$ 5

Voltage, $V=6V$

To find: The ratio of power consumed in the combination of resistors in each case, P_S/P_R

Solution: (i) Series case

In series, total resistance is given as-

$R_S=R1+R2$

$R_S=15+15$

$R_S=30\Omega$ (1)

Thus, the total resistance in series, R_S is 30Ω

Now, We know that, power consumed by a resistor is given as-

Power=Voltage^2 / Resistance

Or, $P=V^2/R$

Substituting the value of V and R

we get- $P_S=6^2/30$

$P_S=36/30$

$P_S=1.2W$ (1)

Thus, the power consumed in series, P_S is 1.2 watt.

(ii) Parallel case:

In parallel, total resistance is given as-

$1/R_P=1/R1+1/R2$

$1/R_P=1/15+1/15$

$1/R_P=2/15$

$R_P=15/2$

$R_P=7.5\Omega$ (1)

Thus, the effective resistance in parallel, R_P is 7.5Ω

Now,We know that, power consumed by a resistor is given as-

Power=Voltage^2 /Resistance

Substituting the value of V and R

we get-$P_R=6^2/7.5$

$P_R=36/7.5$

$P_R=360/75$

$P_R=4.8W$ (1)

Thus, the power consumed in parallel, P_R is 4.8 watt.

Ratio of power consumed in the two combinations,

$P_S / P_R=1.2\ W / 4.8\ W$

$=1/4=1:4$

Thus, the ratio of power consumed in the combination of resistors in each case is (1)

OR

B. (i) Joule's Law of Heating states that the amount of heat produced in a conductor is directly proportional to the square of the electric current passing through it, the resistance of the conductor, and the time for which the current flows. Mathematically, it can be expressed as (2)

$$H = I^2Rt$$

- *H* is the heat produced (in joules),
- *I* is the electric current (in amperes),
- *R* is the resistance of the conductor (in ohms),
- *t* is the time for which the current flows (in seconds).

(ii) Let the equivalent resistance in series be denoted by R_s and that for parallel be denoted by R_p. Total voltage of the circuit is given by V in both cases and the time is denoted by t.

- $R_S = 2+4 = 6\Omega$ (1)
- $1/R_P = 1/2 + 1/4 = 3/4$, $R_P = 4/3$ (1)
- $H_S = V^2 t/R_S$, $H_P = V^2 t/R_P$
- $H_S/H_P = R_P/R_S = 2/9$ (1)

SECTION – E		
37	A. The primary ore of mercury is Cinnabar, which is found in nature as Mercury(II) sulfide (HgS). (1) B. When zinc carbonate is heated in absence of air, carbon dioxide and zinc oxide are formed.Calcination is the process that involves the reduction of carbonate ores to oxides in the absence of oxygen. (1) C. The metal X that reacts with ferric oxide (Fe_2O_3) in a highly exothermic reaction to join railway tracks is aluminum (Al).The chemical equation for the reaction is: Fe_2O_3 (s) + 2 Al (s) → 2 Fe (l) + Al_2O_3 (s) + Heat This reaction is also known as the thermit reaction. It's a displacement reaction where aluminum displaces iron from iron oxide. The reaction produces a large amount of heat that melts the iron, which is then used to join railway tracks or machine parts. (2) - *Student to attempt either subpart C or D.* D. Carbon cannot reduce sodium oxide to sodium because sodium is more reactive than carbon and has a greater affinity for oxygen. Sodium is a highly reactive metal and is placed at the top of the reactivity series. Carbon is below sodium in the reactivity series, so it is not as reactive. Sodium can be obtained from sodium chloride through electrolytic reduction. During this process, molten sodium chloride is electrolyzed, and sodium is deposited at the cathode while chlorine is liberated at the anode: Cathode: $Na^+ + e^- \rightarrow Na$ Anode: $2Cl^- \rightarrow Cl_2 + 2e-$ (2)	4
38	A. Convex lens. (1) B. Converging property. The lens can converge parallel rays to one point. (1) **Student to attempt either subpart C or D** C. To correct hypermetropia, lenses of telescopes, microscopes and slide Projectors (2) **OR** D. If the object is kept between the optical centre and the focus the image obtained is virtual, rest in all cases the image is real. (2)	
39	Generally there are 6 type of tropism namely phototropism, gravitropism (geotropism), chemotropism, thigmotropism, thermotropism and hydrotropism. (2) **Student to attempt either subpart A or B** A. It is an example of thigmotropism. Abscisic acid (2) B. Growth of pollen tubes towards the egg cell is one example of chemotropism (1) C. Cytokinins promotes cell division in plants. (1)	